

2 (**long**) Figure 3 shows a field effect transistor (FET) configured in an amplifier circuit. The impedances of C_{in} , C_s and C_{out} are negligible at mid-band frequencies. The circuit is biased so that $V_{GS} = -2$ V, $V_{DS} = 10$ V and $I_{DS} = 1$ mA. At this operating point the small signal parameters of the FET are $g_m = 10$ mA V⁻¹ and $r_d = 25$ k Ω .

- (a) Explain the purposes of the capacitors C_{in} , C_s and C_{out} in the circuit. [2]
- (b) Calculate the values for R_1 and R_2 to achieve the stated operating point. [6]
- (c) For the amplifier circuit of Fig. 3 draw the small signal model for mid-band frequencies. Hence calculate the small signal voltage gain, and the input and output impedances. [12]
- (d) A load of resistance $R_L = 10$ k Ω is connected across the output terminals. Assuming that the lower 3 dB cut-off frequency of the amplifier is dominated by the effect of C_{out} , what value of C_{out} is required if the circuit is to have a 3 dB low-frequency cut-off of 100 Hz? [6]
- (e) The previously stated value of the external load resistor $R_L = 10$ k Ω does not maximise the power in the load. Calculate a new value of R_L in order to maximise the signal power in the load. [4]

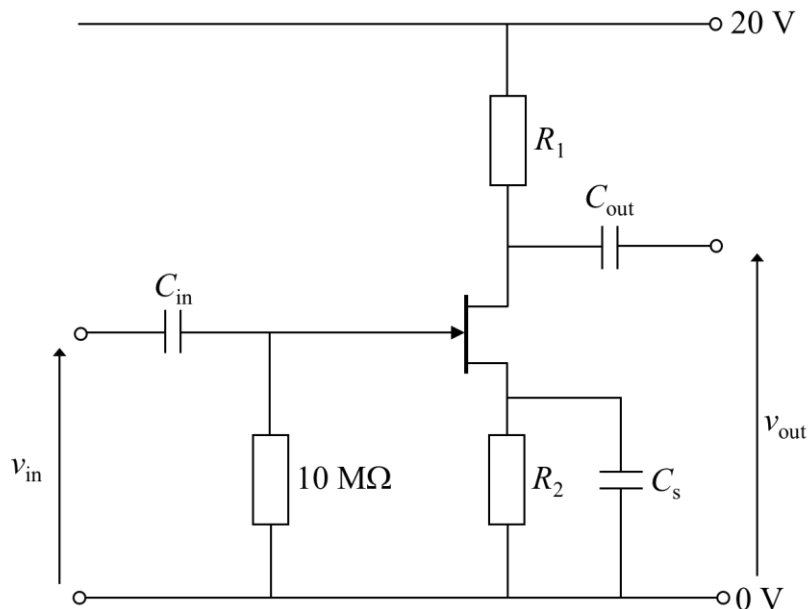


Fig. 3